

PowerPoint Presentation: Chapter 11 - Influenza Vaccination: Seasonal & Pandemic Preparedness

Long-term Protection Beyond Pneumococcal Vaccines

Slide 1: Strategic Positioning in Module

Building on Chapter 10: Pneumococcal Focus

Previous Chapter Highlights:

- PCV13 science & age-specific schedules
- T-cell dependent immunity vs polysaccharide
- Special populations & implementation
- Indian context adaptations

Chapter 11 focuses on influenza - the other major pneumonia vaccine:

- **Seasonal influenza burden** and prevention
 - **Vaccine types & immunogenicity**
 - **Pandemic preparedness frameworks**
 - **Healthcare worker vaccination ethics**
 - **Public health program integration**
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Slide 2: Chapter Learning Objectives

By session end, students will be able to:

1. Quantify seasonal influenza pneumonia burden globally & in India
2. Differentiate influenza vaccine types and their indications
3. Apply evidence-based seasonal vaccination recommendations
4. Design pandemic influenza response strategies
5. Counsel patients on influenza vaccination benefits and timing

CBME Competencies: Medical Knowledge, Practice-Based Learning, Systems-Based Practice

Slide 3: Viral Pneumonia's Second Major Threat

Influenza as bacterial pneumonia's enabler

- **Primary viral infection** (influenza A/B viruses)
- **Destruction of respiratory epithelium**
- **Secondary bacterial invasion** (*S. pneumoniae*, *S. aureus*)
- **10-15% of influenza cases develop pneumonia**
- **High-risk populations:** Elderly, immunocompromised, pregnant

Dual viral-bacterial pathogenesis diagram

Slide 4: Global Influenza Burden & Economic Impact

Annual seasonal influenza statistics:

- **300-500,000 deaths worldwide** (including pneumonia complications)
- **3-5 million severe illness cases** requiring hospitalization
- **Economic burden: \$87 billion USD** annual global costs
- **India-specific:** 10-20 million cases, 50,000-100,000 deaths annually

World map showing influenza burden by WHO region

Slide 5: Influenza Virus Characteristics

Family Orthomyxoviridae - distinctive features

Type A (most virulent):

- Hemagglutinin (HA): 18 subtypes (H1-H18)
- Neuraminidase (NA): 11 subtypes (N1-N11)
- Pandemic potential through antigenic shift
- Examples: H1N1 (2009 pandemic), seasonal H3N2

Type B (seasonal):

- No subtypes, two lineages: Victoria & Yamagata
- Seasonal epidemics, no pandemics
- Less severe than type A
- Antigenic drift requires annual vaccine updates

Virus structure and surface antigen illustrations

Slide 6: Antigenic Evolution - Annual Vaccine Challenge

Why influenza vaccines need yearly reformulation

Antigenic Drift:

- Point mutations in HA/NA genes
- Gradual immune escape from prior vaccines
- Predictable seasonal circulation patterns
- Vaccine mismatch risk (20-50% annually)

Antigenic Shift:

- Reassortment between human/animal viruses
- Sudden emergence of novel subtypes
- Pandemic potential (H5N1, H7N9 surveillance)

- Rapid vaccine development required
- Evolution timeline showing drift vs shift

Slide 7: Influenza Vaccine Types - Comparative Analysis

Vaccine Type	Administration	Age Range	Composition	Efficiency
LAIV (Nasal spray)	Intranasal	2-49 years	Live attenuated	50-60%
IIV (Standard dose)	IM injection	≥6 months	Inactivated	40-50%
HD-IIV (High dose)	IM injection	≥65 years	4× antigen	55-70%
RIV (Recombinant)	IM injection	≥18 years	Egg-free	45-55%
QIV (Quadrivalent)	IM injection	≥6 months	4 strains	50-65%

Differentiated vaccine comparison chart

Slide 8: Seasonal Timing & Geographical Variations

Northern Hemisphere Timing:

- **Primary season:** December-March (peaking January-February)
- **Vaccine optimization:** September-October pre-season
- **Community protection duration:** 6-8 months
- **Booster consideration:** High-risk individuals

Southern Hemisphere Timing:

- **Primary season:** April-September (peaking June-August)
- **Vaccine optimization:** March-April preparation
- **Strain prediction:** Basis for Northern vaccine design
- **Global influenza surveillance:** Continuous year-round

Dual hemisphere vaccination calendar

Slide 9: Priority Groups for Seasonal Vaccination

Universal Recommendation Categories:

1. **All persons ≥6 months** - Community protection
2. **Pregnant individuals** - Maternal & fetal protection
3. **Children 6-59 months** - Primary high-risk age group
4. **Adults ≥65 years** - Immunosenescence vulnerability
5. **Chronic medical conditions** - COPD, diabetes, heart disease
6. **Healthcare workers** - Patient protection and occupational health
7. **Residents of long-term care facilities** - Outbreak prevention

Priority population pyramid with percentages

Slide 10: Vaccine Effectiveness in Real-World Settings

Seasonal Influenza Vaccine Effectiveness:

- **Healthy adults (18-64 years):** 40-60% against symptomatic illness
- **Elderly (≥ 65 years):** 50-70% against severe outcomes
- **Children (2-17 years):** 60-75% protection demonstrated
- **High-dose formulation:** 25% additional protection in elderly

Secondary Pneumonia Prevention:

- **Hospitalization reduction:** 30-50% in vaccinated groups
- **ICU admission decline:** 40-60% fewer severe cases
- **Mortality benefit:** 80% reduction in influenza-related deaths

Effectiveness data with confidence intervals

Slide 11: Pediatric Vaccination Special Considerations

Children 6 months to 8 years - unique requirements

First Vaccination Ever:

- **Two doses required:** 4 weeks apart for optimal response
- **Immune naive individuals:** Primary series needed
- **Subsequent years:** Annual single dose
- **LAIV option:** Ages 2-8 years (better immunogenicity)

Catch-up Situations:

- **6-35 months:** 0.25 mL dose (standard pediatric formulation)
 - **36+ months:** Adult dose (0.5 mL)
 - **Historical barriers:** Needle fear, parental concerns
 - **School-based programs:** High coverage achievement potential
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Slide 12: Pregnancy & Maternal Immunization Benefits

Universal Recommendation: All trimesters safe and effective

Maternal Benefits:

- Severe influenza prevention during pregnancy
- Reduced hospitalization and complications
- Postpartum immunity maintenance

Fetal/Infant Benefits:

- Placental antibody transfer until 6 months
- Neonatal protection against severe illness
- Reduced infant hospitalization (31-63%)
- Extended protection through breastfeeding

Safety Profile:

- Extensive safety data from millions of doses
 - No increased adverse pregnancy outcomes
 - Category B classification (FDA pregnancy categories)
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Slide 13: Immunocompromised Patient Considerations

HIV-Infected Individuals:

- Safe regardless of CD4 count (>200 recommended)
- Better response with CD4 >500 cells/ μ L
- Combination with pneumococcal vaccination advised
- Annual revaccination recommended

Cancer Patients:

- Timing with chemotherapy cycles crucial
- Safe during remission or stable disease
- May need higher antibody titers post-vaccination
- Often requires annual vaccination

Organ Transplant Recipients:

- Post-transplant protection (3-6 months recommended)
 - May require higher antigen doses
 - Reduced response in immunosuppressive states
 - Live vaccines (LAIV) generally contraindicated
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Slide 14: Adverse Events and Safety Profile

Common Local Reactions (10-30%):

- **Injection site pain, redness, swelling**
- **Generally mild, resolve within 24-48 hours**
- **No specific treatment usually required**
- **Higher incidence with high-dose formulations**

Systemic Reactions (5-15%):

- **Low-grade fever, fatigue, myalgia**
- **Headache, arthralgia, chills**
- **Generally self-limiting within 3-7 days**
- **More common in younger adults than elderly**

Rare Serious Events:

- **Guillain-Barré syndrome:** Not causally linked (<1:1 million doses)**
 - **Anaphylaxis:** Extremely rare (1:1 million)**
 - **New onset thrombocytopenia:** Idiopathic, coincidental**
 - **Oculorespiratory syndrome:** Rare, self-limiting**
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Slide 15: LAIV (Live Attenuated) Special Considerations

Advantages over inactivated vaccines:

- **Better mucosal immunity** (intranasal administration)
- **Longer duration of protection**
- **Cross-protection** against drifted viruses
- **Easier administration** (needle-free)

Disadvantages & Contraindications:

- **Not recommended in pregnancy**
- **Contraindicated in immunocompromised**
- **Avoid in children with asthma** (risk of wheezing)
- **Garlic/onion allergy** (stabilizer component)
- **Recent influenza antiviral use** (reduces efficacy)

Effectiveness Data:

- 50-58% against antigenically matched viruses
 - Better for children 2-17 years
 - Lower effectiveness in adults >60 years
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Slide 16: Pandemic Preparedness - From Seasonal to Global Threat

Pandemic Influenza Characteristics:

- **Rapid global spread** ($R_0=2.0-3.0$)
- **Novel viral strains** (limited population immunity)
- **High healthcare system strain**
- **Economic disruption unparalleled**

Pandemic impact timeline (2009 H1N1 example)

Slide 17: Pandemic Vaccine Development Strategies

Traditional Egg-Based Production:

- **6-9 months development timeline**
- **Stockpiling of egg-adapted strains**
- **Global distribution challenges**

- **Limitations in low/middle-income countries**

Next-Generation Platforms:

Cell-Based Vaccines:

- **Faster production cycles** (weeks instead of months)
- **No egg allergies risk**
- **Higher fidelity to circulating viruses**
- **Scalable manufacturing capacity**

RNA/DNA Vaccines:

- **24-48 hour development** possible
 - **Highly adaptable** to any genetic sequence
 - **Stored in freeze-dried form** (long shelf-life)
 - **Pandemic-ready stockpiles** available
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Slide 18: Healthcare Worker Vaccination Ethics

Professional Responsibility Framework:

Patient Protection:

- **Institutional safety** through herd immunity
- **Prevention of nosocomial transmission**
- **Workplace absenteeism reduction**
- **Quality patient care continuity**

Personal Health Protection:

- **Occupational hazard** in healthcare settings
- **2009 H1N1 pandemic lessons** (high HCW mortality)
- **Work-life balance** through illness prevention
- **Medication compliance incentives**

Ethical Component:

- **Duty of care** to vulnerable patients
 - **Benedict's principle** (first, do no harm)
 - **Institutional policies** vs individual autonomy
 - **Mandate effectiveness** in high-coverage settings
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Slide 19: Vaccine Hesitancy in Influenza Context

Influenza-Specific Concerns:

"It's just the flu - not serious"

- *Response:* Seasonal influenza causes 290,000-650,000 deaths annually

"Vaccine causes the flu"

- *Response:* Inactivated vaccines cannot cause infection

"I never get the flu"

- *Response:* 5-20% of adults get symptomatic influenza annually

"Too many side effects"

- *Response:* Local reactions mild and transient (<2% serious)

"Not worth it for pregnant women"

- *Response:* Prevents maternal-fetal complications, safe in all trimesters

Evidence-based rebuttal framework

Slide 20: Cost-Effectiveness Analysis

Economic Justification for Universal Influenza Vaccination

Direct Cost Savings:

- **Outpatient visits reduction:** 40-60% fewer consultations
- **Hospitalization prevention:** 30-50% reduction in severe cases
- **Antibiotic prescription decline:** 25-40% fewer prescriptions
- **Work productivity gains:** 15-25% absenteeism reduction

Cost per QALY:

- **Developed countries:** \$10,000-30,000 per QALY gained
- **Developing countries:** Even more favorable economics
- **Universal coverage ROI:** 2-5 fold return on investment
- **Societal perspective:** 6-8 months to break even per vaccine

Economic evaluation scatter plot

Slide 21: Global Repurposing for COVID-19 Response

Influenza Program Infrastructure for Pandemic Response

Established Surveillance Networks:

- **WHO FluNet:** Global viral surveillance system
- **EMFLU:** Emerging respiratory virus monitoring
- **GISRs:** Global Influenza Surveillance States
- **Real-time genomic sequencing**

Vaccine Manufacturing Capacity:

- **Northern Hemisphere factories:** Rapid strain switching

- **Southern Hemisphere positioning:** Year-round production
 - **Novel platform technologies:** mRNA vaccine adaptability
 - **Stockpile management:** Strategic reserve implementation
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Slide 22: Future Directions - Universal Influenza Vaccines

Beyond Seasonal Vaccines:

Conserved Epitope Vaccines:

- **Hemagglutinin stalk** rather than head antigens
- **Universal protection** against all influenza A subtypes
- **Potent cross-protection** against drifted viruses
- **Single-dose lifetime immunity** theoretical possibility

Mucosal Administration:

- **Intranasal delivery** for superior local immunity
- **Mucosal adjuvants** to enhance response
- **Easier administration** for mass campaigns
- **No injection anxiety** barriers

Combination Vaccines:

- **Single-shot protection** (influenza + pneumococcal)
 - **COVID-19 integration** for respiratory protection
 - **Adjuvanted formulations** for older adults
 - **Personalized vaccination** by genetic risk factors
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Slide 23: CBME Key Assessment Questions

Question: Influenza vaccine effectiveness in elderly?

- A) 20-30%
- B) 50-70% against severe outcomes ★
- C) 80-90%
- D) 0-10%

Question: LAIV effectiveness in children vs adults?

- A) Better in adults
- B) Equal effectiveness
- C) Better in children (>60%) ★
- D) Not effective in either

Question: Pregnancy influenza vaccination timing?

- A) Only first trimester
- B) Only second trimester

- C) All trimesters safe ★
 - D) Only postpartum
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Slide 24: Clinical Decision-Making Scenarios

Case 1: 75-year-old COPD Patient

- History: Frequent exacerbations, recent hospitalization
- Recommendation: High-dose quadrivalent influenza vaccine
- Rationale: Enhanced immunogenicity in elderly + COPD risk
- Additional: Pneumococcal vaccination for comprehensive protection

Case 2: 32-year-old Pregnant Healthcare Worker

- Trimester: Second trimester, multiple patient contacts
 - Recommendation: Standard quadrivalent IIV
 - Rationale: Fetal protection + occupational risk reduction
 - Counseling: Safety profile, benefit-to-risk assessment
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Slide 25: Session Evaluation & Q&A

Learning Experience Assessment:

What was most valuable today? _____

What needs more emphasis? _____

Additional topics you'd like covered? _____

Overall Rating: ★ ★ ★ ★ ★

Interactive Discussion Topics:

- Hospital policy on mandatory HCW vaccination
 - Challenges in rural immunization delivery
 - Integration with existing respiratory illness programs
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Slide 26: Practical Implementation Strategies

Clinic-Level Integration:

- **Annual campaigns:** September-November focused activities
- **Standing order protocols:** Nurse-initiated vaccination
- **Reminder systems:** Automated patient messaging
- **Documentation:** Integration with medical records

Community Approaches:

- **Pharmacy partnerships:** Extended access hours

- **Workplace vaccination:** Employer-sponsored programs
 - **Senior center campaigns:** Concentrated vulnerable populations
 - **Door-to-door services:** Hard-to-reach geographic areas
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Slide 27: References & Resources

Key Guidelines (2024 Updates):

1. **ACIP Influenza Vaccination Recommendations**
2. **WHO Global Influenza Strategy 2019-2030**
3. **CDC Pink Book - Influenza Chapter**
4. **IAP Pediatric Immunization Schedule India**

Online Platforms:

- **CDC Influenza:** <https://www.cdc.gov/flu>
 - **WHO FluNet:** <https://www.who.int/tools/flunet>
 - **Influenza Research Database:** Comprehensive literature
 - **VIA - Vaccine Information for Adults:** ACIP resource
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Slide 28: Next Session: Program Implementation (Chapter 12)

Building on Influenza Foundation:

- **Coverage monitoring systems**
- **Social mobilization strategies**
- **Cold chain logistics excellence**
- **Quality assurance frameworks**
- **Digital transformation in immunization**

Key Takeaways from Today's Session:

- **Vaccination prevents 50-80% of influenza pneumonia cases**
 - **Annual updates required due to viral evolution**
 - **Universal recommendation benefits all age groups**
 - **Pregnancy safe and effective for mother-baby protection**
 - **Healthcare workers have ethical vaccination responsibility**
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Summary Slide Title: Module Chapter 11: Influenza Vaccination Essentials

Total Slides: 28 **Estimated Duration:** 55-65 minutes **Faculty:** Dr. H S Lingaiah, MD, FIAHS **Target:** MBBS students mastering respiratory infection prevention